UNIT V

Drone flying and operations

A 60-minute interactive flight-operations session

Plan, choose modes, control, fly safely, monitor, comply

Today’s 60-minute flight plan

0–10	10–20	20–32	32–42	42–52	52–60
ConOps and risk	Flight modes and failsafes	Dynamics and control	Safe operations	Logs and monitoring	DGCA rules and exit ticket

Learning outcome

Plan a safe small-UAV operation and justify its flight modes, control settings, limits and permissions.

A ConOps answers six questions before anyone flies

WHAT

Mission, users and the data or service delivered

WITH WHAT

Aircraft, payload, software version, flight modes allowed

WHERE

Indoor or outdoor, obstacles, people, airspace zone

WHO

Remote pilot, visual observer, payload operator

LIMITS

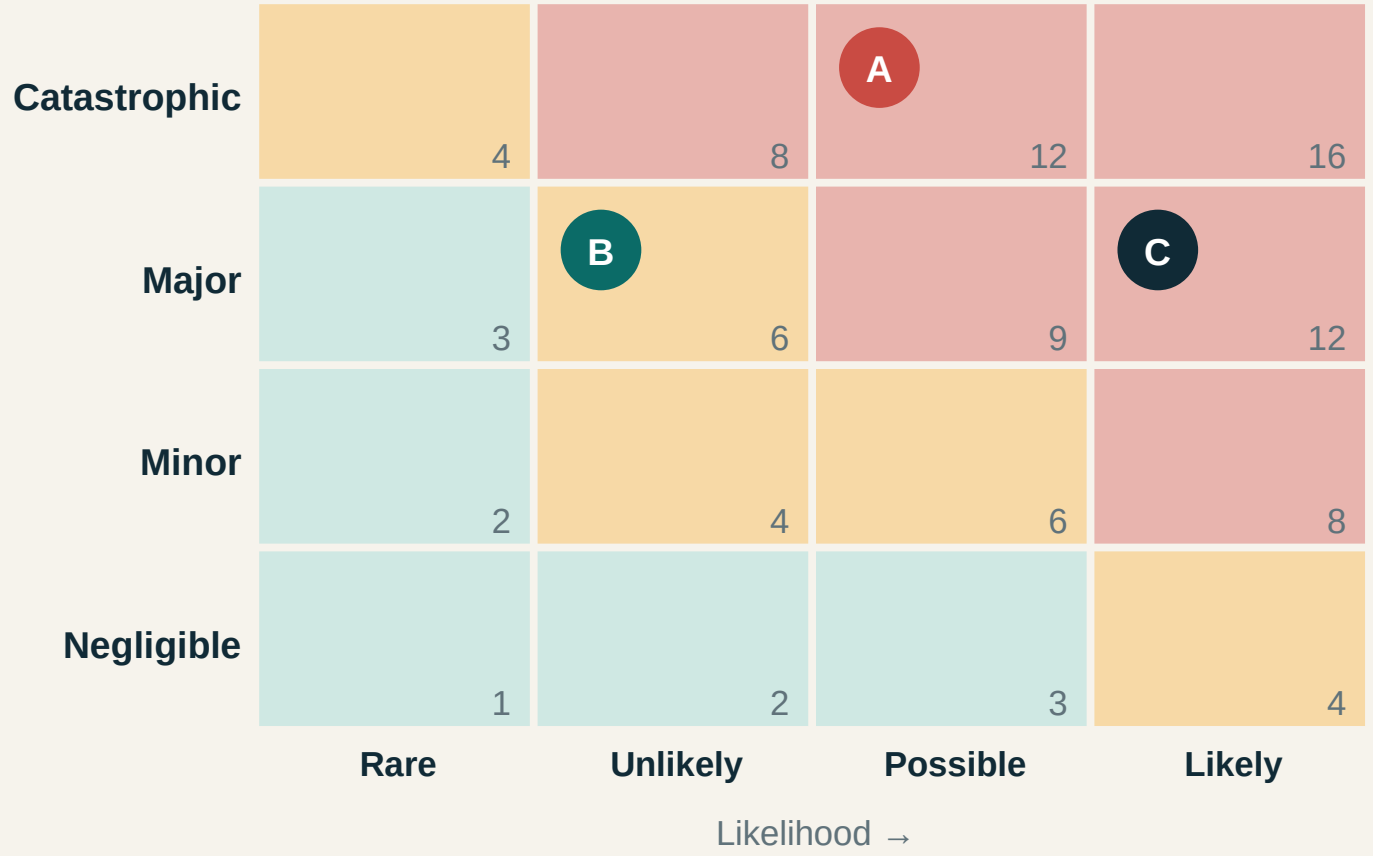
Altitude, distance, wind, battery thresholds

WHAT IF

Failures, planned responses, emergency landing area

Scenario: indoor inspection of a warehouse roof truss with a 250 g micro-quad. Fill the six cards in 4 minutes.

Risk = severity × likelihood; a control moves the dot



Hazards on the board

- **A** Fly-away over a crowd
- **B** Battery fire during charging
- **C** Propeller strike on a crew member

A control lowers likelihood (geofence, pre-flight checks) or severity (cordon, prop guards). Record the residual risk.

PANEL: DRAG ONE HAZARD TO A NEW CELL

Each flight mode trades pilot workload for sensor dependence

	SENSORS NEEDED	PILOT COMMANDS
Acro / Rate	Gyro	Rates
Stabilize	+ accelerometer	Attitude
AltHold	+ barometer	Climb rate
Loiter	+ GNSS, compass	Velocity
Auto / Mission	Full navigation	Supervise
RTL / Land	Valid home + nav	Abort / override

Rule of thumb

Every step up adds a sensor.

Every added sensor is a new way to fail.

Know which mode to fall back to when it does.

ASK: WHAT FAILS FIRST IN LOITER?

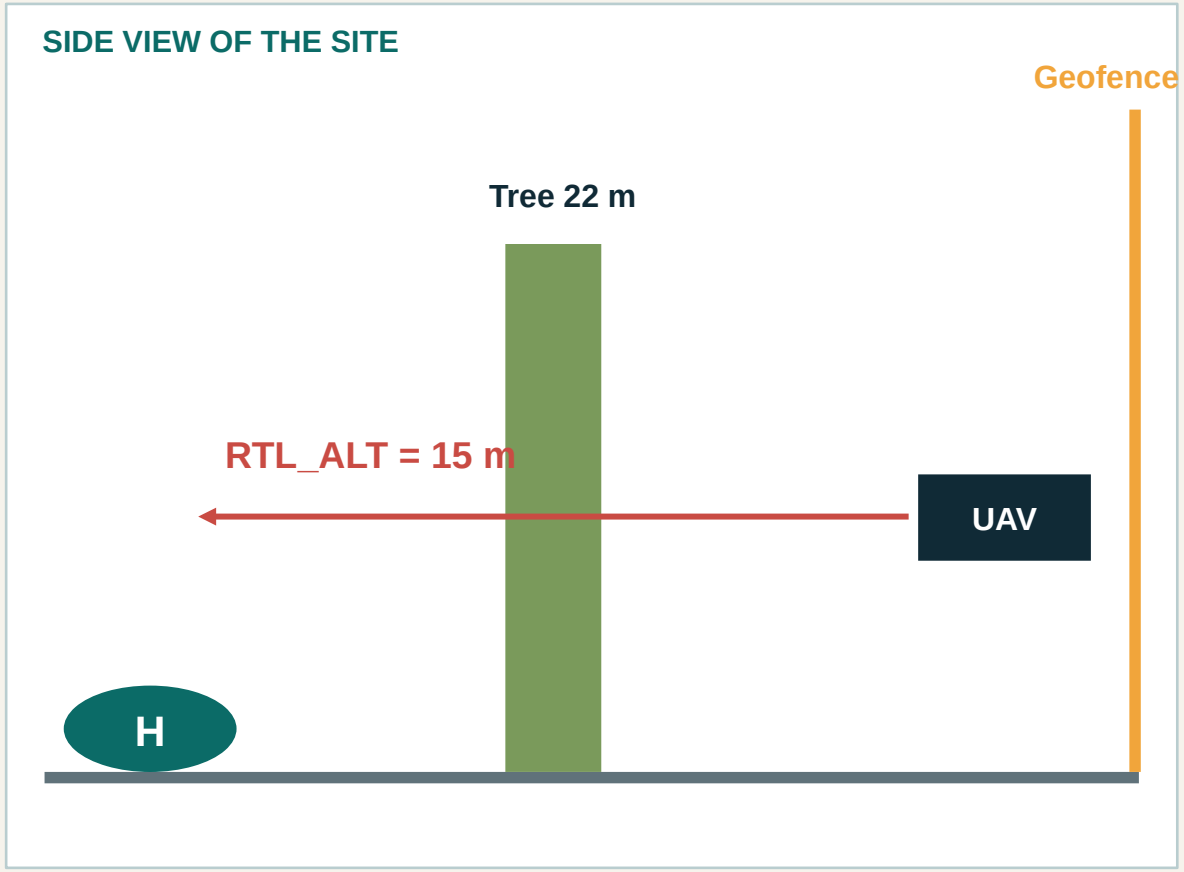
Return-to-launch is only as safe as its four checks

1. **Home**
   Recorded at arming with a good GNSS fix

2. **Altitude**
   RTL altitude above every obstacle on the way home

3. **Energy**
   Enough to return from the farthest point into the wind

4. **Estimate**
   Healthy heading and position (EKF)



Fix it: raise RTL altitude, re-route home, or plan a LAND failsafe.

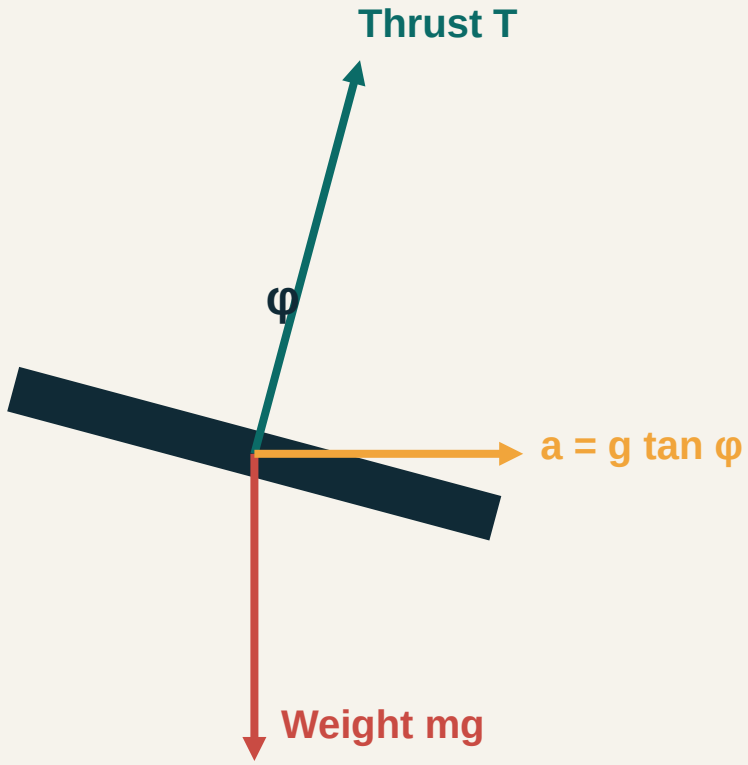
Failsafes are resolved by hazard severity

CRITICAL	Critical battery · attitude estimate lost · motor failure	**LAND now**
CONTAINMENT	Geofence breach	**Hold or RTL**
LINK	RC or telemetry lost	**RTL (or continue a planned mission)**
ADVISORY	Low battery warning · degraded GNSS	**Alert pilot, plan recovery**

Higher rows override lower rows. Write the priority table before the flight, not during it.

ASK: BATTERY CRITICAL AND OUTSIDE THE FENCE?

A multirotor moves by tilting its thrust



Hold altitude while tilted

$T \cos \varphi = mg \rightarrow T = mg / \cos \varphi$

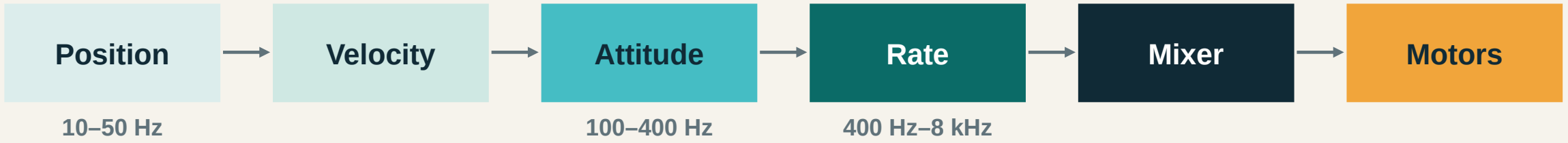
$T \sin \varphi = ma \rightarrow a = g \tan \varphi$

At $\varphi = 15°$: $a = 2.63\ m/s^2$, $T = 1.035\ mg$

CHECK

What tilt gives 5 m/s² of forward acceleration, and how much extra thrust does it need?

Cascaded loops: every inner loop runs faster



State estimator (EKF): IMU + barometer + GNSS + compass feed every loop

- Each loop turns its error into a set-point for the loop inside it.
- An inner loop must be about 3–10× faster than the loop around it, so the outer loop sees it as ideal.
- Flight modes enter this chain at different points: Acro at Rate, Stabilize at Attitude, Loiter at Position.

ASK: WHY TUNE THE RATE LOOP FIRST?

P, I and D each fix one problem and create another

Raise	Rise time	Overshoot	Steady-state error	Side effect
Kp	Faster	More	Smaller	Oscillation if too high
Ki	Faster	More	Removed (step)	Windup, slow oscillation
Kd	Small change	Less	No effect	Amplifies noise, hot motors

Windup

The integrator keeps charging while the motors are saturated, then overshoots. Fix: stop integrating while saturated (clamping).

PREDICT

Response is sluggish, has low overshoot and a steady error. Which gain do you change first, and what will you watch in the log?

PANEL: PREDICT, THEN SKETCH THE STEP RESPONSE

Safe flying runs on roles, numbers and checklists

Remote pilot	In command; may abort and take manual control at any time
Visual observer	Watches aircraft, airspace and people; warns, never flies
Payload / GCS operator	Runs camera and mission so the pilot can fly

Go/no-go item	Example limit
Wind / gusts	≤ 8 m/s / ≤ 10 m/s
Visibility	VLOS throughout
GNSS	Pre-arm checks pass
Battery	Full; IR near baseline
Crew	Briefed and rested

Checklist cycle

Pre-flight → **In-flight** → **Post-flight** → **Emergency**

Wind over the limit means no flight. Not a short hop.

ASK: WHO CAN ABORT THE MISSION?

Plan the return with ground speed, not airspeed

Worked example

Out and back 2 km at 12 m/s airspeed, 4 m/s headwind on the way out.

Out: 12 − 4 = 8 m/s → 2000/8 = 250 s

Back: 12 + 4 = 16 m/s → 2000/16 = 125 s

Total: 375 s, not the 333 s of still air

The wind can turn. Plan the return as if it were into the wind, then add the landing reserve.

YOUR TURN

Same aircraft, 1.5 km out, 6 m/s headwind on the return leg.

1. Outbound and return times?

2. With 9 min of usable flight time and a 25% reserve, is it a go?

Logs tell the story: align time, then follow the chain



Whatever rose first is the most likely cause. Vibration first points to a propeller, motor or mount.

Firmware	Onboard log	Ground log
ArduPilot	.bin (DataFlash)	.tlog
PX4	.ulg (ULog)	—

Tools: UAV Log Viewer, PX4 Flight Review, Mission Planner, PlotJuggler.

Signals to plot

- Desired vs estimated altitude
- Throttle and motor outputs
- Vibration and clipping
- Battery voltage and current

Before you fly in India: category, UIN, RPC, zone

Nano	Micro	Small	Medium	Large
≤ 250 g	250 g – 2 kg	2 – 25 kg	25 – 150 kg	> 150 kg

GREEN	Up to 400 ft AGL where not red or yellow	**No permission needed**
YELLOW	Controlled airspace, including near airports	**ATC permission**
RED	No-drone zones: airports, borders, strategic sites	**Central Government only**

- UIN on Digital Sky for each drone; type certificate unless exempt.
- RPC from a DGCA-authorised RPTO; not needed for nano, or micro flown non-commercially. Check the Digital Sky map every flight.

ASK: DOES A 1.5 KG CLUB DRONE NEED AN RPC?

EXIT TICKET

One site. One mode plan. One reason.

1. **Name the flight mode you would fall back to if GNSS failed, and why.**

2. **State the RTL altitude for a site with a 22 m tree, with your margin.**

3. **Give one quantitative go/no-go limit and what you do if it is exceeded.**

Write directly on the panel or answer in 30 seconds with your partner.

CO5 - Apply